Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/07/2005
PI Name:	Keil, David G. Ph.D.		
Project Title:	Particle Generation and Evolution in Silane/Acetylene Flames in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	08543-2229	**Congressional District:**	12
Comments:			
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
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Flight Program:			
Flight Assignment:			
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Grant/Contract No.:	NAS3-01092		
Performance Goal No.:			
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Task Description:	To better understand particle formation processes and their coupling with flame propagation processes through detailed characterization of microgravity flames that produce copious amounts of particulate 'smoke'. The effects of silane/acetylene ratio and inert diluents on the burning velocities, flame temperatures, particle radiation, and silicon carbide (SiC) particle generation processes in premixed silane/acetylene flames will be measured in microgravity environments using flame diagnostics and analysis of flame products. These microgravity measurements will provide flame data uncomplicated by flame buoyancy and particle settling and thereby provide the opportunity to elucidate the couplings among particle formation, combustion enthalpy release/transport, and flame propagation. Comparisons of stoichiometric flames producing silicon carbide particles with acetylene-rich flames largely producing carbonaceous soot will advance the understanding of these processes and lead to better means for controlling them.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
	The planned approach to generating and studying premixed silane/acetylene 'vee' and/or cone flames supported above a nozzle burner during a 2.2 second drop tower experiment had been tested in laboratory experiments earlier in the program. The approach involves using a fixed inert gas flow to fill a bladder in a reservoir containing the premixed silane/acetylene reactants, and thereby feeding the reactants to the burner at the identical inert flow rate by displacement. This unconventional approach is required because of the limited drop time in which to stabilize separate reactant burner flows to produce a constant stoichiometric flame and because the total quantity of pyrophoric silane on board must be limited for overall safety reasons. Safety requires that both the pyrophoric silane reactant and the flammable hydrogen flame product be protected from air exposure. Hence, the flame chamber can not be vented to the atmosphere, but rather must be sealed. Constant pressure operation then requires a rapid and reliable mechanism to compensate for the inflow of reactant gases and the expansion of hot flame product gases. Additional inert gas flow is necessary to dilute the hydrogen product beyond its flammability limits. The approach adopted to maintaining constant pressure is to vent any excess gas into an evacuated chamber through a high flow back pressure regulator. The back pressure regulator can maintain a constant average pressure, but a limited response rate generates some pressure fluctuations. Laboratory tests confirmed that proper sequencing can minimize the effect of such pressure excursions on the flame response and stability. These tests were made using surrogate methane/air flames which have comparable burning velocities to those anticipated for the silane/acetylene flames. Some differences in flame response (to pressure variations) may arise due to differing transport properties of the reactants, but such differences in flame response are not expected to prevent flame stabilization in microgravity experiments.
Task Progress:	The above approach was shown to be feasible, and design requirements were defined. The rig design was challenging due to the requirement of providing a completely contained, high flow rate, yet constant pressure, flame environment. Assembly of the experimental rig based on the completed design accepted in a review meeting at the NASA Glenn facility is nearly complete. An application for the required safety permit has been submitted, and the rig is expected to receive permits for use in both the 5 sec drop tower facility, and the 2.2 second tower. This will provide the rapid turn around times available with the 2.2 second tower, as well as the longer experimental times provided by the 5 second tower. Initial experiments are planned to be performed in the 2.2 second tower. If longer observation times are required under specific experimental conditions, the 5 second tower will be used. A no-cost extension in the performance period will be requested to facilitate completion of the planned experiments. In preparation for analysis and interpretation of microgravity flame measurements, probable flame reaction mechanisms have been investigated using available data on silicon-carbon-hydrogen species energetics and reaction rate constants, as well as measured burning velocities and flame temperatures from our laboratory. This analysis has indicated that particle formation is the dominant energy-releasing process that generates the high flame temperature and drives the combustion wave. Considerable gas-phase pre-particle chemistry can occur rapidly at lower temperatures generating little net energy release. This suggests a simplified approach to modeling and gaining insight into the detailed flame chemistry. The approximation assumes energy from the high temperature particle growth region is transported toward the cool reactants and generates a hot bath gas with a fixed temperature profile in which most of the gas phase reactions occur nearly isothermally (at the local temperature). This approximation is currently being tested with kinetic modeling approaches.
Bibliography Type:	Description: (Last Updated: 02/27/2004)